Energy changes

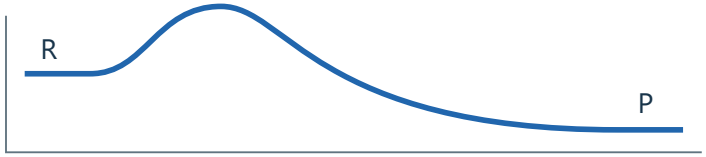
Track energy

1 Exothermic vs endothermic

Exothermic: energy to surroundings; temperature rises; products lower in energy. Combustion, many oxidations, neutralisation.

Endothermic: energy from surroundings; temperature falls; products higher. Thermal decomposition is an example.

2 Reaction profiles



Exothermic: products (P) lower

Vertical axis: energy. Horizontal: reaction progress, not time. Activation energy = minimum energy for a successful collision.

Draw a curved peak. Activation energy: reactant level → peak. Overall change: reactant → product level.

Measure & calculate

3 Bonds: break minus make

Breaking bonds takes energy; forming bonds releases energy. Overall = total broken − total formed.

Negative → exothermic; positive → endothermic. Count every bond and equation coefficient; use supplied values.

4 Temperature practical

Insulated cup/lid, measured volumes, same starting temperature. Mix/stir; record maximum/minimum. Change = extreme − initial.

Vary one factor; control others, repeat and average. Heat loss can reduce measured rise; use time-graph extrapolation when requested.

Generate electricity

5 Cells & batteries

Different electrodes + electrolyte → electricity. Voltage depends on both; compare under controlled conditions.

Non-rechargeable stops when a reactant runs out. External current reverses reactions in rechargeable cells. Series cell voltages add.

6 Hydrogen fuel cells

Continuous H_2 and O_2 supply; hydrogen oxidised, oxygen reduced. Overall: **$2H_2 + O_2 \rightarrow 2H_2O$.**

Electrons go through external load; ions through electrolyte.

Compare refuelling/recharging, storage, cost, lifetime and whole-process emissions, not only water at the outlet.